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FILE

Labels and Annotations: Title Subtitle X-Label Y-Label Z-Label Legend Colorbar Grid X-Grid Y-Grid

Files

Name

- e1t1.m
- e1t2.m
- e1t3.m
- e2t1.m
- e2t2.m
- e2t3.m
- e3t1.m
- exp1_tc1.m
- exp1_tc2.m
- exp1_tc3.m

Workspace

Name	Value	Size	Class
d	1x50 double	1x50	double
f	[900,3500,2...	1x3	double
Freq	[900,3500,2...	1x3	double
Loss	[91.5249,10...	1x3	double
PL1	1x50 double	1x50	double
PL2	1x50 double	1x50	double
PL3	1x50 double	1x50	double

```

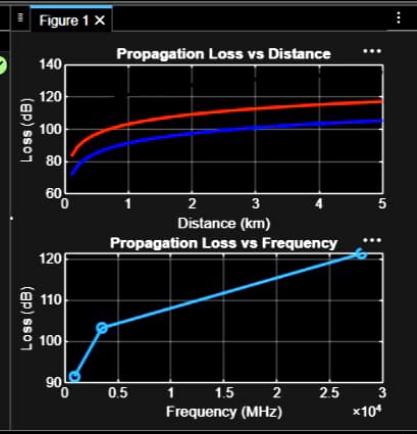
MATLAB Drive/e3t1.m
7 PL3 = 32.44 + 20*log10(T(3)) + 4;
8 figure
9 % ----- Graph 1 -----
10 subplot(2,1,1)
11 plot(d,PL1,'b','LineWidth',2); hold on
12 plot(d,PL2,'r','LineWidth',2)
13 plot(d,PL3,'k','LineWidth',2)
14 title('Propagation Loss vs Distance (km)')
15 xlabel('Distance (km)')
16 ylabel('Loss (dB)')
17 grid on
18 % ----- Graph 2 (NO legend used) -----
19 subplot(2,1,2)
20 Freq = [900 3500 28000];
21 Loss = [PL1(10) PL2(10) PL3(10)]
22 plot(Freq,Loss,'-o','LineWidth',2)
23 title('Propagation Loss vs Frequency (MHz)')
24 xlabel('Frequency (MHz)')
25 ylabel('Loss (dB)')

```

Command Window

New to MATLAB? See resources for [Getting Started](#).

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Debugger

Breakpoints

- ☐ Pause on Errors
- ☐ Pause on Warnings
- ☐ Pause on NaN or Inf
- ☐ Pause on Output

Function call stack

Not paused in debugger

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Name

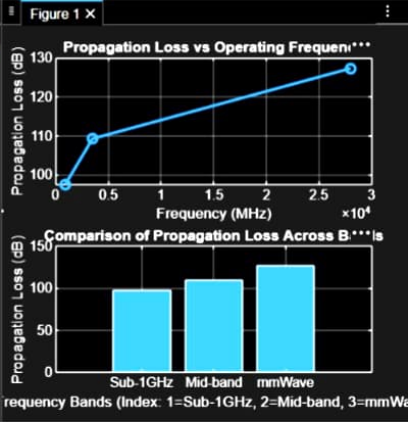
e1t1.m e1t2.m e1t3.m e2t1.m e2t2.m e2t3.m e3t1.m e3t2.m exp1_tc1.m exp1_tc2.m

Workspace

Name	Value	Size	Class
d	2	1x1	double
f	[900,3500,2...]	1x3	double
PL	[97.5455,10...]	1x3	double

```

1  clc; clear; close all;
2  d = 2; % fixed distance (km)
3  f = [900 3500 28000]; % MHz
4  PL = 32.44 + 20*log10(f) + 20*log10(d);
5  figure
6  % ----- Graph 1: Frequency vs Propagation Loss
7  subplot(2,1,1)
8  plot(f,PL,'-o','LineWidth',2)
9  title('Propagation Loss vs Operating Frequency')
10 xlabel('Frequency (MHz)')
11 ylabel('Propagation Loss (dB)')
12 grid on
13 % ----- Graph 2: Band Comparison
14 subplot(2,1,2)
15 bar(PL)
16 title('Comparison of Propagation Loss Across Frequency Bands')
17 xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
18 ylabel('Propagation Loss (dB)')
19 xticklabels({'Sub-1GHz','Mid-band','mmWave'})
    
```



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Pause on Errors Pause on Warnings Pause on NaN or Inf Pause on Output

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matlab.mathworks.com

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Copy FigurePrintShow Code

FILE

MATLAB Drive

Files

Name

e1t1.m

e1t2.m

e1t3.m

e2t1.m

e2t2.m

e2t3.m

e3t1.m

e3t2.m

e3t3.m

exp1_tc1.m

Workspace

Name	Value	Size	Class
d	2	1×1	double
f	[900,3500,2...	1×3	double
PL	[97.5455,10...	1×3	double

mXe3t1.mXe3t2.mXe3t3.m

/MATLAB Drive/e3t3.m

```
3 f = [900 3500 28000];
4 % Distance (km)
5 d = 2;
6 % Propagation Loss (FSPL)
7 PL = 32.44 + 20*log10(f) + 20*log10(d);
8 figure
9 % ----- Graph 1: Simple Line
10 subplot(2,1,1)
11 plot(f, PL, '-o', 'LineWidth', 2)
12 title('Propagation Loss vs Operating Frequency')
13 xlabel('Frequency (MHz)')
14 ylabel('Propagation Loss (dB)')
15 grid on
16 % ----- Graph 2: Safe Comparison
17 subplot(2,1,2)
18 plot(1:3, PL, '-s', 'LineWidth', 2)
19 title('Band-wise Propagation Loss Comparison')
20 xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
21 ylabel('Propagation Loss (dB)')
```

Figure 1 X

Propagation Loss vs Operating Frequency



Frequency (MHz)	Propagation Loss (dB)
900	97.5455
3500	107.5455
28000	127.5455

Propagation Loss (dB)

Frequency (MHz) ×10⁴

Band-wise Propagation Loss Comparison



Band Index	Propagation Loss (dB)
1 (Sub1GHz)	97.5455
2 (Mid-band)	107.5455
3 (mmWave)	127.5455

Propagation Loss (dB)

Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)

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